

Sophia Bian

650-391-3758 | sophiabian777@gmail.com | [linkedin.com/in/bian-sophia](https://www.linkedin.com/in/bian-sophia)

EDUCATION

NORTHWESTERN UNIVERSITY

Evanston, IL

B.S. in Electrical Engineering, Minor in Computer Science

Expected June 2028

- Engineering Dean's Honors List (2025, 2026) | GPA: 3.98/4.00

TECHNICAL SKILLS

Languages: C, C++, Python, MATLAB

Hardware: Altium Designer (schematic capture, multilayer layout), NI Multisim, SMD soldering & rework

Embedded: STM32 (ARM Cortex-M), CAN, SPI, ADC, PWM | **Other:** SOLIDWORKS, Onshape, Git

EXPERIENCE

NORTHWESTERN FORMULA RACING

Evanston, IL

Embedded Engineer, Data Acquisition

September 2025 – Present

- Lead end-to-end development of 2 unique sensor PCBs: Altium schematic and layout, C/C++ firmware, bench bring-up, and on-car integration
- Enable suspension characterization under real driving loads, giving the team measured data to validate future designs and the corner load feedback required for torque vectoring
- Build Python tooling generating dashboard signal viewers from the team CAN DBC file, used as the source of truth for brake bias and sensor calibration during vehicle integration
- Design the training board and contribute to the onboarding curriculum built around it for new members

ROGERS RESEARCH GROUP

Evanston, IL

Undergraduate Research Assistant, Querrey Simpson Institute for Bioelectronics

March 2026 – Present

- Assemble and functionally verify flexible PCBs for a wireless fluorimeter, diagnosing and repairing failed units to yield working boards for experimental use
- Characterize fluorescence sensor response on the bench across sample media with contrasting optical properties, quantifying unit-to-unit variation across assembled boards

PROJECTS

Raspberry Pi CM4 Vehicle Interface Board | *Altium, C++, Python, CAN, MIPI DSI*

2025 – Present

- Laid out a 4-layer carrier board across 2 revisions in Altium integrating a Raspberry Pi CM4, CAN transceiver, USB 2.0, MIPI DSI video output to the driver display, and driver I/O (addressable LEDs, buttons, inputs)
- Designed the power tree for 3 A nominal with input protection, sizing power traces and polygon pours for the load and impedance-matching the CAN differential pair
- Brought both revisions up from bare board through hand soldering, power rail checkout, firmware flash, and vehicle integration
- Traced intermittent flickering and false colors on the dashboard's addressable LED strips to the PWM drive, and designed AND-gate and multiplexer logic into the rev-2 layout to drive all strips from a single PWM output
- Wrote the C++ application decoding vehicle CAN traffic to display live sensor measurements on the driver display, including a diagnostics page that surfaces inverter and system faults for analysis
- Designing next revision to chain the strips into a single addressable run, eliminating the gating logic and halving the board footprint

Suspension & Brake Monitoring System | *Altium, STM32, C++, CAN, ADC*

2026 – Present

- Designing 4 corner boards forming a distributed sensor network that aggregates pushrod and tie-rod strain gauges, suspension potentiometers, and brake rotor temperature sensors, conditioning each signal path and publishing readings to the vehicle CAN network
- Designing the analog front ends for each sensor type, selecting amplifier topology and filter response per signal path, and reworking the strain channel to cut zero-drift error measured on last year's boards
- Spec'ing the full BOM including MCU and power topology against a 38.5 mm × 21 mm target envelope for corner mounting, then hand-assembling and bringing up each board

(E)Education Board | *Altium, STM32, C++, CAN, ADC*

2026 – Present

- Designing a reference board (STM32, CAN transceiver, input protection, external ADC) intended as the schematic template for the subteam's future sensor boards
- Scoping the board's feature set to drive the subteam's onboarding path, exercising SMD soldering, CAN communication, firmware debug, and Altium schematic-to-layout in a single build